

✓ Data Analytics III

Loading the required libraries

```
# Import required libraries
import pandas as pd
import numpy as np
```

Importing the datasetA

```
df= pd.read_csv('4.Iris.csv')
```

Performing operations on dataset

```
df
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Setosa
1	2	4.9	3.0	1.4	0.2	Setosa
2	3	4.7	3.2	1.3	0.2	Setosa
3	4	4.6	3.1	1.5	0.2	Setosa
4	5	5.0	3.6	1.4	0.2	Setosa
...	...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3	Versicolour
146	147	6.3	2.5	5.0	1.9	Versicolour

```
df.shape
```

```
(150, 6)
```

```
df.describe()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.762238
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

```
df.isnull()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	
0	False	False	False	False	False	
1	False	False	False	False	False	
2	False	False	False	False	False	
3	False	False	False	False	False	
4	False	False	False	False	False	
...	...	...	...	...	...	
145	False	False	False	False	False	
146	False	False	False	False	False	
147	False	False	False	False	False	
148	False	False	False	False	False	
149	False	False	False	False	False	

150 rows × 6 columns

```
df.isnull().sum()
```

	0
Id	0
SepalLengthCm	0
SepalWidthCm	0
PetalLengthCm	0
PetalWidthCm	0
Species	0

dtype: int64

```
x= df.drop(["Species"],axis=1)
y = df["Species"]
```

```
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.3)

print(x_train.shape)
print(y_train.shape)
print(x_test.shape)
print(y_test.shape)
```

(120, 5)
(120,)
(30, 5)
(30,)

Train Naive Bayes Classifier Model

```
from sklearn.naive_bayes import MultinomialNB
```

```
classifier= MultinomialNB()
classifier.fit(x_train,y_train)
```

▼ MultinomialNB ⓘ ?

MultinomialNB()

```
classifier.score(x_test,y_test)
```

0.8333333333333334

```
y_pred= classifier.predict(x_test)
```

y_pred

```
array(['Iris-virginica', 'Iris-versicolor', 'Iris-setosa',
      'Iris-virginica', 'Iris-setosa', 'Iris-versicolor', 'Iris-
setosa',
      'Iris-versicolor', 'Iris-versicolor', 'Iris-versicolor',
      'Iris-virginica', 'Iris-versicolor', 'Iris-versicolor',
      'Iris-versicolor', 'Iris-versicolor', 'Iris-setosa',
      'Iris-versicolor', 'Iris-virginica', 'Iris-versicolor',
      'Iris-setosa', 'Iris-virginica', 'Iris-versicolor', 'Iris-
setosa',
      'Iris-setosa', 'Iris-virginica', 'Iris-setosa', 'Iris-
setosa',
      'Iris-virginica', 'Iris-virginica', 'Iris-setosa'],
      dtype='<U15')
```

y_test

	Species
114	Iris-virginica
62	Iris-versicolor
33	Iris-setosa
107	Iris-virginica
7	Iris-setosa
100	Iris-virginica
40	Iris-setosa
86	Iris-versicolor
76	Iris-versicolor
71	Iris-versicolor
134	Iris-virginica
51	Iris-versicolor
73	Iris-versicolor
54	Iris-versicolor
63	Iris-versicolor
37	Iris-setosa
78	Iris-versicolor
90	Iris-versicolor

```

45    Iris-setosa
16    Iris-setosa
121   Iris-virginica
66    Iris-versicolor
24    Iris-setosa
8     Iris-setosa
126   Iris-virginica
22    Iris-setosa
44    Iris-setosa
97    Iris-versicolor
93    Iris-versicolor
26    Iris-setosa

```

dtype: object

Confusion Matrix

```

import sklearn.metrics

lbs = ['Iris-versicolor', 'Iris-setosa', 'Iris-virginica']
print(sklearn.metrics.confusion_matrix(y_test, y_pred, labels = lbs)

```

```

[[10  0  3]
 [ 1 10  0]
 [ 1  0  5]]

```

```

from sklearn.metrics import classification_report

```

```

from sklearn.naive_bayes import GaussianNB
gnb = GaussianNB()
gnb.fit(x_train, y_train)

```

▼ GaussianNB ⓘ ?

GaussianNB()

```
classifier.score(x_test,y_test)
```

```
0.8333333333333334
```

```
y_pred=gnb.predict(x_test)
```

```
y_pred
```

```
array(['Iris-virginica', 'Iris-versicolor', 'Iris-setosa',
       'Iris-virginica', 'Iris-setosa', 'Iris-virginica', 'Iris-
       setosa',
       'Iris-versicolor', 'Iris-versicolor', 'Iris-versicolor',
       'Iris-virginica', 'Iris-versicolor', 'Iris-versicolor',
       'Iris-versicolor', 'Iris-versicolor', 'Iris-setosa',
       'Iris-versicolor', 'Iris-versicolor', 'Iris-setosa', 'Iris-
       setosa',
       'Iris-virginica', 'Iris-versicolor', 'Iris-setosa', 'Iris-
       setosa',
       'Iris-virginica', 'Iris-setosa', 'Iris-setosa', 'Iris-
       versicolor',
       'Iris-versicolor', 'Iris-setosa'], dtype='<U15')
```

```
y_test
```

	Species
114	Iris-virginica
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134	Iris-virginica
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73	Iris-versicolor
..	..

```
54 Iris-versicolor
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```

dtype: object

